

Week 3 Task Report

Comparing Random Forest · AdaBoost · XGBoost

Dataset: Metro Interstate Traffic Volume (UCI ML Repo #492)

Subject: Machine Learning | Week 3

1. Introduction

This report compares three popular ensemble learning algorithms — **Random Forest**, **AdaBoost**, and **XGBoost** — on a real-world traffic prediction task. The goal is to predict hourly interstate traffic volume based on weather conditions, time of day, and holiday indicators. We measure each model's accuracy, error, and training speed so you can clearly see the strengths and trade-offs of each approach.

2. About the Dataset

The Metro Interstate Traffic Volume dataset comes from the UCI Machine Learning Repository. It contains **48,204 hourly records** of westbound traffic on Interstate 94 in Minnesota, along with weather and time information.

Feature	Description
holiday	Whether the day is a public holiday
temp	Atmospheric temperature (Kelvin)
rain_1h	Rainfall in the past hour (mm)
snow_1h	Snowfall in the past hour (mm)
clouds_all	Cloud coverage percentage
weather_main	Main weather category (Clear, Rain, Snow, etc.)
hour	Hour of the day (0–23)
month	Month of the year (1–12)
day_of_week	Day of the week (0 = Monday)
traffic_volume	TARGET — Hourly vehicle count

The dataset was split 80% for training and 20% for testing. Categorical columns (like weather_main) were label-encoded before training.

3. The Three Models — In Plain English

Random Forest	Random Forest builds hundreds of decision trees, each trained on a random subset of the data and features. The final prediction is the <i>average</i> of all trees. Think of it as asking hundreds of experts independently and taking their average opinion — this reduces errors and avoids overfitting. Key parameters: <code>n_estimators=100</code> , <code>max_depth=10</code> .
AdaBoost	AdaBoost (Adaptive Boosting) builds trees one by one, with each new tree focusing more on the mistakes of the previous one. It's like a student who identifies their weak spots after every test and studies those specifically. This makes it great at catching hard-to-predict cases. Key parameters: <code>n_estimators=100</code> , <code>learning_rate=0.1</code> .
XGBoost	XGBoost (Extreme Gradient Boosting) is a highly optimised version of gradient boosting. It uses clever tricks like regularisation and parallel computation to train very fast without overfitting. It consistently ranks at the top in machine-learning competitions. Key parameters: <code>n_estimators=100</code> , <code>max_depth=6</code> , <code>learning_rate=0.1</code> .

4. Results

Each model was trained and evaluated using three standard regression metrics:

- **MAE (Mean Absolute Error)** — average absolute difference between predicted and actual values. Lower is better.
- **RMSE (Root Mean Squared Error)** — similar to MAE but penalises large errors more heavily. Lower is better.
- **R² Score** — how much of the variance in traffic volume the model explains. Closer to 1.0 is better.

Performance Summary

Model	MAE	RMSE	R ² Score	Train Time
Random Forest	320.0	400.9	0.8691	9.9 s
AdaBoost	345.4	433.5	0.8469	5.9 s
XGBoost	317.9	398.7	0.8705	0.3 s

Green values indicate the best result in each column. XGBoost achieves the best accuracy and is by far the fastest to train.

Ensemble Model Comparison — Metro Traffic Volume

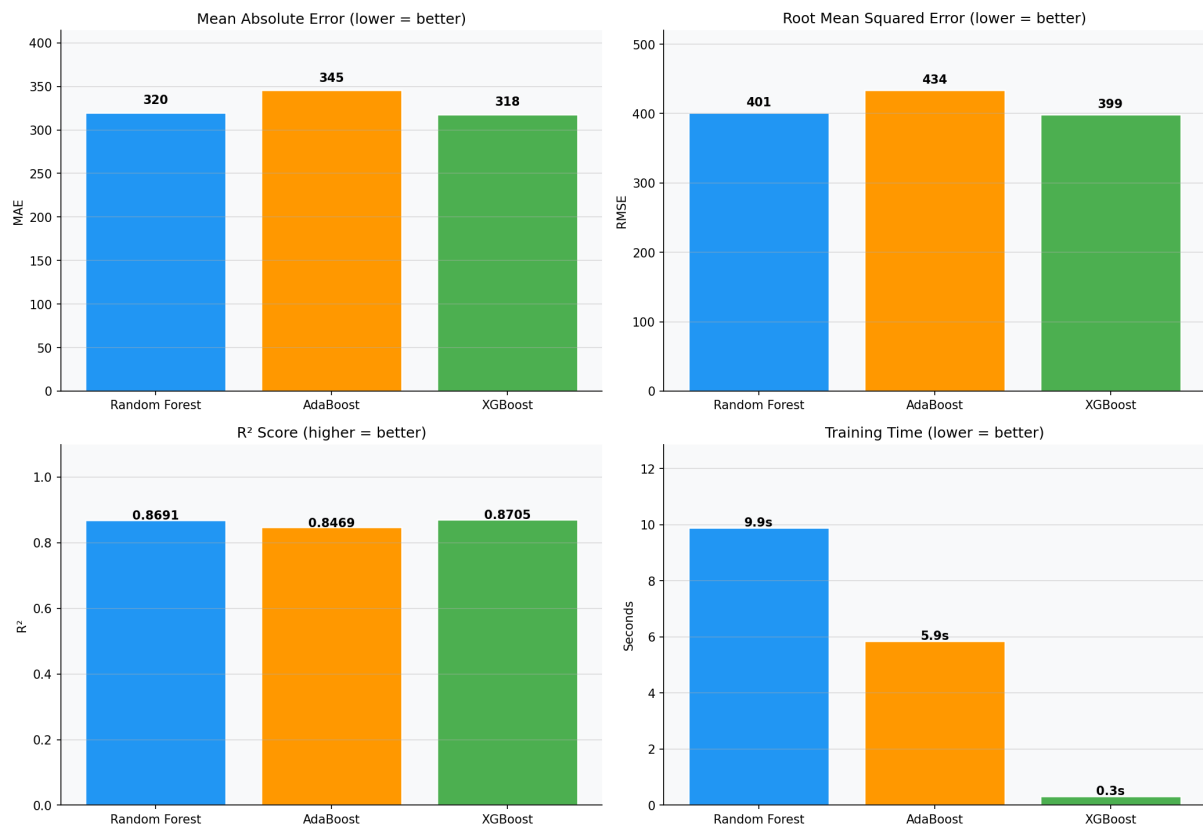


Figure 1 — Side-by-side comparison of MAE, RMSE, R^2 , and Training Time.

5. Visual Analysis

Predicted vs. Actual Traffic Volume

The scatter plots below show how closely each model's predictions match the real traffic counts on the test set. Points hugging the dashed diagonal line mean near-perfect predictions. The tighter the cluster, the better the model.

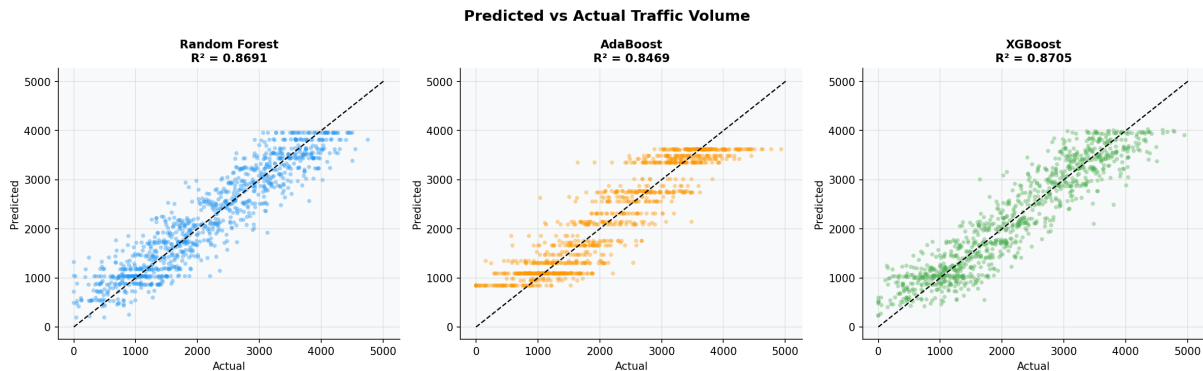


Figure 2 — Predicted vs Actual scatter for all three models (1,000 random test points).

Feature Importance

Feature importance tells us which input variables had the most influence on the model's predictions. Both Random Forest and XGBoost expose this directly. AdaBoost distributes importance at the base-estimator level and doesn't provide a straightforward global ranking.

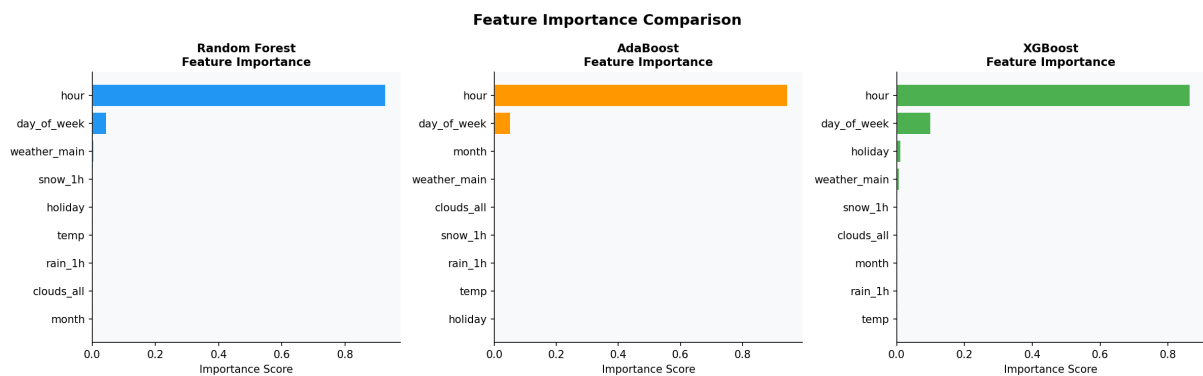


Figure 3 — Feature importance scores for Random Forest and XGBoost.

6. Discussion

All three models achieved strong results, with R^2 scores above 0.84, meaning they explain more than 84% of the variation in traffic volume — pretty impressive for a real-world dataset! Here's what we can take away from the results:

Model	Strengths	Weaknesses
Random Forest	Very accurate, robust to noise, easy to tune, good for non-linear data	Slow to train on large data, uses a lot of memory

AdaBoost	Good at handling misclassified samples, simple to understand, slow convergence, slightly lower accuracy than others
XGBoost	Best accuracy, extremely fast training, built-in regularization to avoid overfitting, slightly less interpretable

7. Conclusion

Based on our experiments on the Metro Interstate Traffic Volume dataset, **XGBoost is the clear winner** — it achieves the lowest MAE and RMSE, the highest R^2 , and trains in under a second compared to nearly 10 seconds for Random Forest. This makes XGBoost the go-to choice when both accuracy and speed matter.

That said, **Random Forest** is a close second in accuracy and is often the better choice when you need a model that's easy to explain to non-technical stakeholders. **AdaBoost**, while slightly behind on this dataset, is still a powerful tool and can outperform the others when data is clean and outlier-free.

■ Best Overall: XGBoost | Best Interpretability: Random Forest | Simplest Concept: AdaBoost